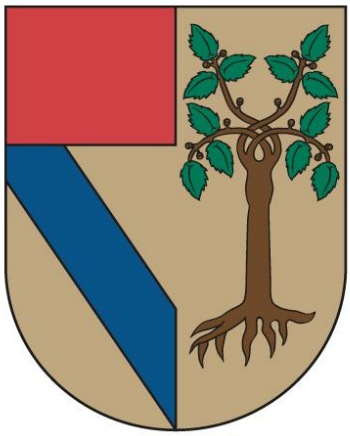


UNIVERSIDAD
PANAMERICANA

Artificial intelligence in videogames

2nd Partial Project

Student:
<your_name>

Professor:
Alfredo Emmanuel Garcia Falcon

Due date:
Friday October 23rd

Objective

The objective of the project is to implement a shortest path search program using the Dijkstra algorithm.

Project description

The project will consist of implementing the Dijkstra shortest path search algorithm. Within the program you must be able to create graphs to be processed by the algorithm with their respective nodes and connections.

The team will be able to decide how to create these graphs, either on a console or visually in a graphics engine. The nodes must have a label to distinguish them and their connections must have the respective weights.

Within the program the graph will be created with its nodes and connections, the Dijkstra algorithm will be executed and at the end the shortest path of the graph will be shown.

Methodology

The project will be developed in teams of 3 people using the programming language decided by the team members.

Project Rubric

The project will consist of two parts: the implementation and execution of the code and the report. The implementation and execution of the code will be 60% of the project grade and the report will be the remaining 40%.

The report must contain the following points:

- Presentation sheet
- Dijkstra algorithm explanation
- Description of each data structure, classes, methods and attributes utilized in the code
- Instructions to graph creation
- Instructions to run the code
- At least three images of the code execution that contain the execution of the program, the creation of graphs and the result of the algorithm
- Conclusion of each of the team members

The code implementation will be divided into components, and it will be graded if they are implemented in the project. The 60% will be divided among all the components in the rubric and each one of them will be added to obtain the 60%. It will be graded as follows: not implemented, implementation has bugs and correctly implemented. If it is not implemented, it won't add those points to the 60%. If it is implemented but has bugs, only half of the points will be counted. And if it is implemented correctly, all the points corresponding to the component will be added.

Here are the components of the project:

Description	Not implemented	Implementation has bugs	Correctly implemented
Class or data structure that represents the graph			
Class or data structure that represents the node with its label, its connections and its weights			
Indication of initial node			
Indication of final node			
Function or method to calculate the weight of a node			
Function, method or condition to select the node with the lowest weight			
Function, method or condition to change a node from unvisited to visited			
Function, method or condition to update the current node to the node with the least weight			
Functionality to create the node in the UI			
Functionality to add weight to the node in the UI			
Animation of Dijkstra algorithm calculating the possible paths			
The program correctly calculates the total weight of the shortest path and displays it on the screen			
The program correctly calculates the shortest path and displays the sequence of nodes on the screen			
Extra 15%: Double Dijkstra algorithm, one from start and another from end node and create the shortest path where both runs intersect (Minimum 300 nodes).			

Deliverables and due date

Despite it is a team project, each of the team members must deliver a .zip file that contains the PDF report and the necessary files to run the program. The file must have the following naming convention: *NameLastname_AIV_2nd_Partial_Project.zip.* For instance: AlfredoGarcia_AIV_2nd_Partial_Project.zip

The file must be uploaded to Classroom no later than Friday October 23rd at 23:59 p.m.